

## Colombia

### Superintendence of Industry and Commerce

The SIC (Superintendencia de Industria y Comercio) has been developing three projects aimed at improving technological tools and analytical techniques that can help the SIC's investigators and forensic engineers to meet the challenges posed by the large volumes of data produced and consumed by today's markets.

These projects were envisioned as tools that could: enhance the Authority's detection capabilities; increase its investigative efficiency when addressing large volumes of structured and unstructured data; pinpoint suspicious signs, patterns, transactions, and exhibit pricing insights within such data, and better allocate its human resources.

#### Sabueso

Sabueso was created with the objective of harnessing the data on products available in online commerce (web-based portals), allowing a baseline of information to analyze the historical behavior of products that have been or that are or could be under the scrutiny of the SIC (Superintendencia de Industria y Comercio). It is a price monitoring tool aimed at enhancing the authority's inspection, surveillance, and control functions. Sabueso has four types of information-gathering bots, each of these bots was specifically designed to traverse a supermarket's website, simulating a human's online interaction with the store. This is how information about the products available for sale in each store is collected.

To speed up the collection of information, the work is divided into categories and the system coordinates the launching of a robot hive. Each of the members of this hive oversees the collection of a single category and after finishing its task is replaced by a different bot. Sabueso allows investigators to view descriptive statistics automatically for the products filtered in the selection tab. The descriptive statistics displayed depict the behavior over time of product prices and the variations between samples. In addition to this type of statistical analysis, there are also applications such as the identification of patterns that indicate warning signs in other products and markets.

Sabueso was recently implemented during a VAT-free day –where certain groups of goods are free of VAT to encourage household consumption and economic recovery promoted by the government. The tool monitored and captured the prices of appliances, apparel, and sports supplies sold by a large department store. The tool allows dynamic visualization in dashboards and will provide insights into firms' price strategies during VAT-free days.

### Inspector

The highly qualified human talent within the Competition Advocacy Group devotes a significant amount of time to purely operational activities. For example, gathering information on new regulatory projects issued by regulatory entities to analyze later whether such projects may impact free competition. The application developed by the SIC solves the problem by monitoring the regulatory projects published on the websites of state regulators. If changes are detected in the published regulations, the system automatically sends an e-mail to the interested party previously defined to inform about the news.

### Sherlock

Sherlock is a project that seeks to support the SIC's investigators in the identification of signs or patterns that suggest potential anticompetitive behaviors with the data available from public procurement processes.

The first stage of development of the project consisted in the creation of a tool that could facilitate the access of investigators to public data available on the web. The purpose is to aid case handlers in the identification of procurement processes with similar characteristics to the one (s) of interest; the bidders within these processes; the contracts awarded by the same entity to the same contractors; relationships or nexus between legal representatives and relevant participants, among others. This first stage of the tool comprised capturing the data, organizing it, and afterward, providing simple descriptive analytics to the investigators. The software cleans, structures, and transforms the collected data and then loads it into a database within the SIC. It also allows for visualization in dashboards with filters, graphs, and comparisons.

Sherlock's second stage, now in the beta phase of development, involves the automation of the search of the abovementioned signs and patterns within the bulk of public procurement data available online. It is based on machine learning technologies and deep machine learning; it relies on artificial intelligence. This second stage will leverage historical data and will analyze and further diagnose information based on indicators already identified by international organizations like the OECD (Organization for Economic Cooperation and Development).

The project will have the capacity of providing informed red flags. It would automatically categorize a particular piece of data into one of the indicators of collusion widely known. This feature will provide predictions useful for the investigators. This would provide a more dynamic Colombia's computational antitrust tools flow of valuable information for the researchers, in addition to greater descriptive insights from the data.